#científiques

Mar Reguant, investigadora en economia de l'energia

Visita al Thau Sant Cugat

2026

Em presento

Sóc la Mar Reguant.

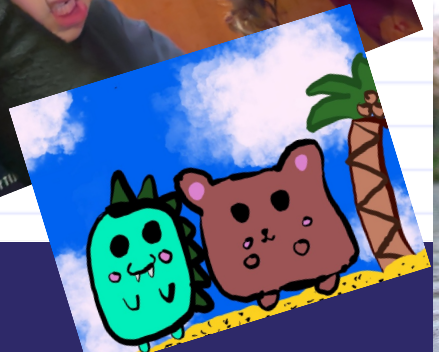
Sóc investigadora en el camp de l'**economia de l'energia**.

Vaig estudia Economia i després vaig fer un doctorat a l'extranger, on hi vaig estudiar i treballar molts anys!

Treballo a un centre del CSIC (Consell Superior d'Investigacions Científiques) a la UAB.

PD: Sóc la mama de la Mel (3r) i el Sam (1r)!

Aficions i temps lliure



Quan tenia la vostra edat...

De gran volia ser...

Aficions...

Quan tenia la vostra edat...

M'agradava...

... la ciència?

... les mates?

... les llengües?

... llegir?

... la gimnàstica?

... l'art i la plàstica?

... la música?

Una mica tot!

Carrera científica

UAB
Universitat Autònoma de Barcelona

IN OMNIBUS VERITAS
UNIVERSITÄT MANNHEIM

Massachusetts Institute of Technology

LELAND STANFORD JUNIOR UNIVERSITY
DIE LUFT DER FREIHEIT WEHT
1891

Toulouse School of Economics

NORTHWESTERN UNIVERSITY
QUAECUMQUE SUNT VERA
1851

#científiques

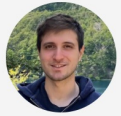
Estic contenta d'haver triat la carrera científica?

Sí

Amb qui treballo?

Mar Reguant
PRINCIPAL INVESTIGATOR
Website

David Andrés-Cerezo
ASSISTANT PROFESSOR
Website

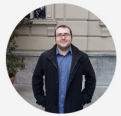


Juan Ignacio Mercatante
POST-DOC
Website

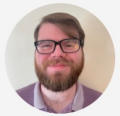
Uxue Arregi
RESEARCH ASSISTANT

Maia Garrell Calsamiglia
RESEARCH ASSISTANT

Mayra Wagner
RESEARCH ASSISTANT



Philipp Steinbrunner
VISITING (SYDDANSK UNIVERISTET)
Website

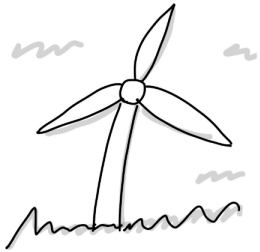


Andrew Hinchberger
VISITING PHD STUDENT (NORTHWESTERN)

Què investigo? / En què treballo?

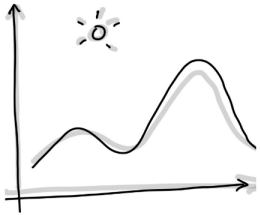
BSE Energy Research Group

Welcome to the research page of the BSE Energy Research Group.



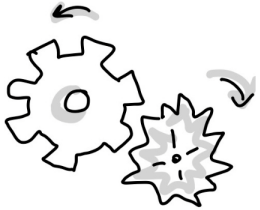
Renewable Power

We analyze the renewable energy boom in the electricity sector. We quantify the benefits and cost...



Demand Side Behavior

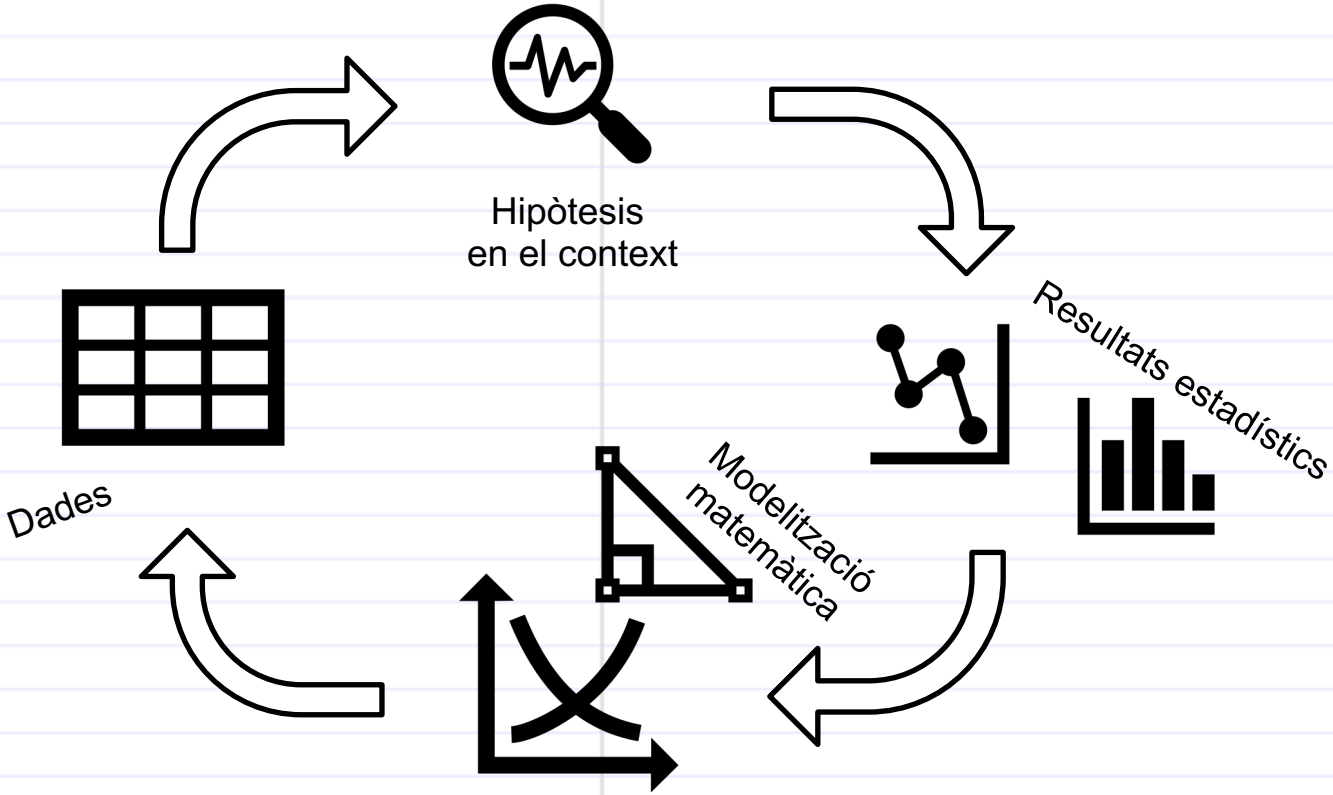
We examine the responses and impact to households from the energy transition. In many of hour pro...



Methods and Tools

Some of our work provides guidance on the methods and tools in the literature of industrial organ...

Com treballo?



Què us explicaré avui?

Com podem fer servir les dades i el codi Python per a entendre **el valor afegit** de les plaques solars del Thau?

Amb quines dades?

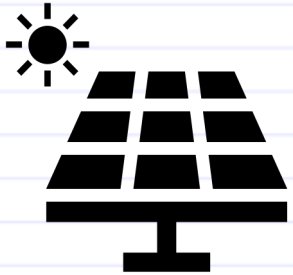
kWh:
Consum elèctric en una hora

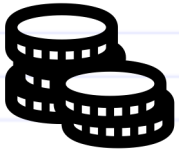
solar_kWh:
Producció solar en una hora

netConsumption:
La diferència, kWh – solar_kWh

month	day	hour	dow	kWh	solar_kWh	netConsumption
1	1	0	5	29	0.0	29.0
1	1	1	5	27	0.0	27.0
1	1	2	5	29	0.0	29.0
1	1	3	5	27	0.0	27.0
1	1	4	5	25	0.0	25.0

Què és el valor afegit?



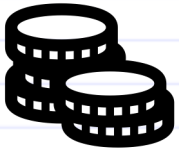


Valor monetari:
Estalvi en la factura

Valor ambiental:
Reducció d'emissions

Les emissions de combustible fòssil generen el 75% del canvi climatic!

I en matemàtiques?



Valor monetari:
Estalvi en la factura

$$EconomicBenefit = \begin{cases} kWhSolar \times price & if \quad kWhSolar < kWh \\ kWh \times price + (kWhSolar - kWh) \times solarPrice & if \quad kWhSolar \geq kWh \end{cases}$$

Valor ambiental:
Reducció d'emissions

$$EnvironmentalBenefit = kWhSolar \times avoided\ emissions \times EUR/ton$$

Anem-hi?

https://mreguant.github.io/public/solarthau.html